C2 | COMPONENT ATLAS (TARGET)

SINGLE-PAGE OVERALL COMPONENT VIEW • 6 OWNED SERVICE BOUNDARIES + FRONTEND/GATEWAY • TARGET COMPONENTS ARE PROPOSED

WEB FRONTEND [React / TypeScript]
Authentication + profile views · Catalog / planning UX · Activity / progress UX · Notification center
Typed API client · Cookie / session handling · Strict HTTPS gateway calls

API GATEWAY [YARP / ASP.NET Core]
Routes + service discovery · JWT edge auth + role policies · Rate limiting · Correlation / forwarding
REST :8080 · Owns NO domain DB · Admin-only operations routing

ROUTING & TRUST BOUNDARY: Browser → Web → Gateway → Service. Each domain service enforces authorization / ownership again. Internal calls use REST or gRPC, never another service's DB.

IDENTITY & PROFILE :8081
Proposed microservice boundary

- **Auth REST** — Register • login • refresh • revoke
- **Profile REST** — Member profile • allergy restrictions
- **Authentication** — Password hash • JWT • claims
- **Session manager** — Rotate/revoke hashed refresh tokens
- **Identity repositories** — Users • profiles • relations
- **Outbox publisher** — ProfileChanged + auth events

Cloudflare D1 refresh adapter optional / OFF

LongevityIdentityDb
OWNED SQL SERVER DATABASE
No cross-service FK

CATALOG & RULES :8082
Proposed microservice boundary

- **Catalog REST** — Foods • recipes • ingredients
- **Rule REST** — Rules • versions • published sets
- **Catalog domain** — Validation • soft delete • indexes
- **Rule governance** — Evidence • sensitive-rule metadata
- **Version publisher** — Immutable approved rule snapshots
- **Outbox publisher** — CatalogChanged • RulePublished

Catalog versions are immutable after publish

LongevityCatalogDb
OWNED SQL SERVER DATABASE
No cross-service FK

PLANNING :8083
Proposed microservice boundary

- **Planning REST** — Generate / revise meal plans
- **Challenge REST** — Daily steps • state transitions
- **FMD safety gate** — Contraindication + consent checks
- **Plan orchestrator** — Scheduling • revision control
- **Recommendation client** — gRPC :8085 → safety ranker
- **Outbox publisher** — PlanGenerated • challenge events

Never bypass medically sensitive safety gate

LongevityPlanningDb
OWNED SQL SERVER DATABASE
No cross-service FK

TRACKING & PROGRESS :8084
Proposed microservice boundary

- **Logging REST** — Meal / activity / eating windows
- **Progress REST** — LDAS + history + summaries
- **Tracking application** — Owner scope + log validation
- **LDAS calculator** — Versioned deterministic dimensions
- **Report projection** — Weekly summaries • event inbox
- **Outbox publisher** — ActivityLogged / progress events

Incomplete source data → partial/NULL LDAS

LongevityTrackingDb
OWNED SQL SERVER DATABASE
No cross-service FK

RECOMMENDATION :8085
Proposed microservice boundary

- **Recommendation gRPC** — Input schema + request validation
- **Hard safety filter** — Allergy / exclusion / FMD checks
- **Deterministic ranker** — Score • reasons • explainability
- **Read-model projector** — Catalog + rule snapshot versions
- **Explanation adapter** — Optional local AI • no safety override
- **Feedback repository** — Record ranker feedback

Live code = gRPC skeleton; target ranker pending

LongevityRecommendationDb
OWNED SQL SERVER DATABASE
No cross-service FK

BACKGROUND WORKER :8086
Proposed microservice boundary

- **Streams consumer** — XREADGROUP • XACK • inbox
- **Reminder scheduler** — Timezone / quiet hours / DST
- **Background executor** — Due reminders • weekly jobs
- **Retry + dead-letter** — Bounded attempts • escalation
- **Ops endpoints** — Readiness + protected admin ops
- **Result publisher** — Worker Outbox → event stream

Implemented reminder path is independently tested

LongevityWorkerDb
OWNED SQL SERVER DATABASE
No cross-service FK

ASYNC EVENT LANE — REDIS STREAMS Per-domain Outbox → XADD → XREADGROUP + XACK → ProcessedEvent/Inbox → Result Outbox / retry / dead letter

KEY SERVICE INTERACTIONS Gateway → Identity/Catalog/Planning/Tracking [REST] • Planning → Recommendation [gRPC] • Worker ↔ Redis [consumer + publisher] • Recommendation ← versioned Catalog/Rule snapshots

IMPLEMENTATION STATUS (2026-11-05) CURRENT: one REST API + TrackingDb + WorkerDb + Redis reminder E2E, gRPC skeleton. TARGET: YARP + six independent domain services and full catalog/planning/ranking. Do NOT mistake proposals for shipped code.

COMPONENTS WITHIN SEPARATE CONTAINERS This one-page atlas intentionally juxtaposes six Level-3 service internals. Official C4 normally gives each container its own C2; this view is a consolidated assignment handout.